Gravity and the Age-old Question

At this point you might think we've reached the end of the bucket story, with Mach's ideas having been discredited and Einstein's radical updating of Newton's absolute conceptions of space and time having won the day. The truth, though, is more subtle and more interesting. But if you're new to the ideas we've covered so far, you may need a break before pressing on to the last sections of this chapter. In Table 3.1 you'll find a summary to refresh your memory when you've geared up to reengage.

Okay. If you're reading these words, I gather you're ready for the next major step in spacetime's story, a step catalyzed in large part by none other than Ernst Mach. Although special relativity, unlike Mach's theory, concludes that even in an otherwise empty universe you would feel pressed against the inside wall of a spinning bucket and that the rope tied between two twirling rocks would pull taut, Einstein remained deeply fascinated by Mach's ideas. He realized, however, that serious consideration of these ideas required significantly extending them. Mach never really specified a mechanism whereby distant stars and other matter in the universe might play a role in how strongly your arms splay outward when you spin or how forcefully you feel pressed against the inner wall of a spinning bucket. Einstein began to suspect that if there were such a mechanism it might have something to do with gravity.

This realization had a particular allure for Einstein because in special relativity, to keep the analysis tractable, he had completely ignored gravity.

Newton	Space is an entity; accelerated motion is not relative; absolutist position.
Leibniz	Space is not an entity; all aspects of motion are relative; relationist position.
Mach	Space is not an entity; accelerated motion is relative to average mass distribution in the universe; relationist position.
Einstein (Special Relativity)	Space and time are individually relative; space-time is an absolute entity.

Table 3.1 A summary of various positions on the nature of space and spacetime.